Regression, correlation and hypothesis testing

1: Exponential Models

Recap of Pure Year 1. Using $y = ab^x$ to model an exponential relationship between two variables.

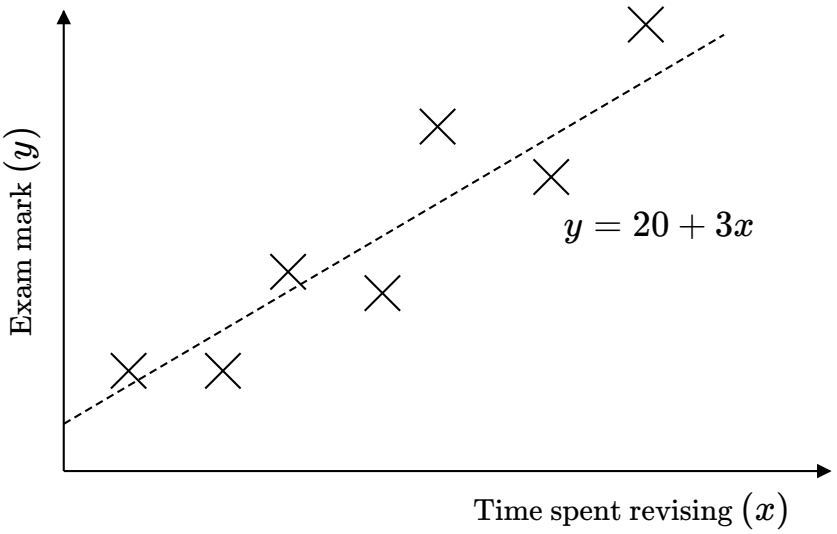
2: Measuring Correlation

Using the Product Moment Correlation Coefficient (PMCC), r, to measure the strength of correlation between two variables.

3: Hypothesis Testing for no correlation

We want to test whether two variables have some kind of correlation, or whether any correlation observed just happened by chance.

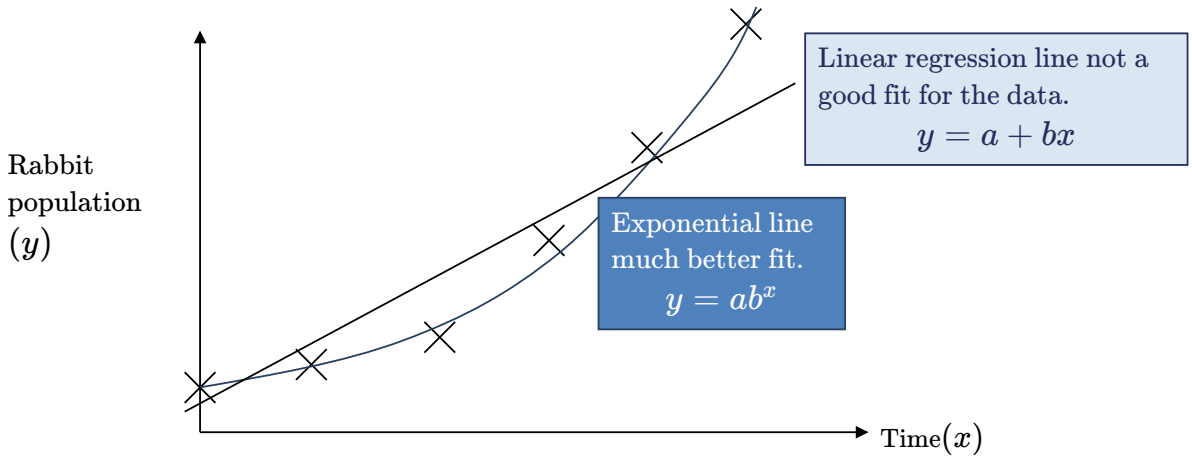
RECAP: What is regression? 3



'Regression' is the act of setting the parameters of our model to best explain the data
(here the parameters are the gradient and y-intercept of the line of best fit)

I record people's exam marks as well as the time they spent revising. I want to predict how well someone will do based on the time they spent revising. How would I do this?

RECAP: Exponential Regression 4



For some variables, e.g. population with time, it may be more appropriate to use an **exponential** equation, i.e. $y = ab^x$, where a and b are constants we need to fix to best match the data.

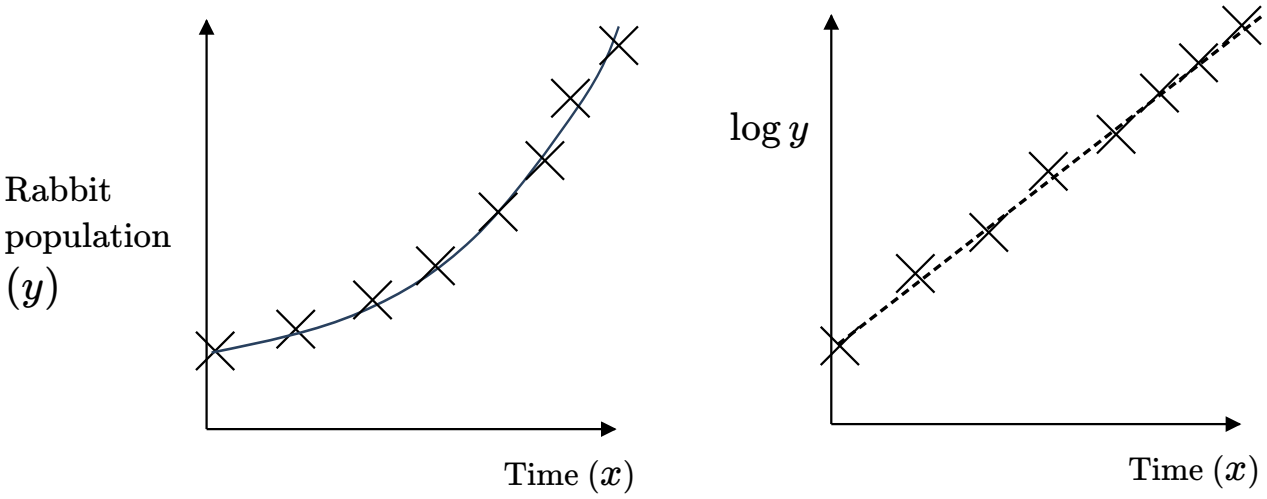
$y = ab^x$

?

If $y = kb^x$ for constants k and b then $\log y = \log k + x \log b$

RECAP: Exponential Regression

If $y = kb^x$ for constants k and b then $\log y = \log k + x \log b$



Comparing the equations, we can see that if we log the y values (although leave the x values), the data then forms a straight line, with y-intercept $\log k$ and gradient $\log b$.

Modelling growth rate against temperature

[Textbook] The table shows some data collected on the temperature, in °C, of a colony of bacteria (t) and its growth rate (g).

Temperature, $\mathbf{t}$ (°C)	3	5	6	8	9	11
Growth rate, g	1.04	1.49	1.79	2.58	3.1	4.46

The data are coded using the changes of variable $x = t$ and $y = \log g$. The regression line of y on x is found to be $y = -0.2215 + 0.0792x$.

a. Mika says that the constant -0.2215 in the regression line means that the colony is shrinking when the temperature is $0°$C. Explain why Mika is wrong

b. Given that the data can be modelled by an equation of the form $g = kb^t$ where k and b are constants, find the values of k and b.

a ?

b ?

Test your Understanding 7

Robert wants to model a rabbit population P with respect to time in years t. He proposes that the population can be modelled using an exponential model: $P = kb^t$
The data is coded using $x = t$ and $y = \log P$. The regression line of y on x is found to be $y = 2 + 0.3x$. Determine the values of k and b.

?

Geogebra Regression model

Power regression on the CG50 8

A website has had the following annual numbers of page views each year:

Year	2012	2013	2014	2015	2016	2017	2018	2019
Views **V**	10115	26790	60306	180386	1119801	8.3 m	21.9 m	*57.5 m*

Determine the appropriate constants if we assume a power model $V = at^b$, where t is the number of years after 2011.

GC50 Instructions

1. *Select statistics mode*
2. *Enter the list (e.g. the ones above where year starts from 1)*
3. *Press the Calc Menu (F2)*
4. *Press the REG menu (F3)*
5. *There are a variety of regressions. Go to Next Page (F6)*
6. *Select Power for power regression*

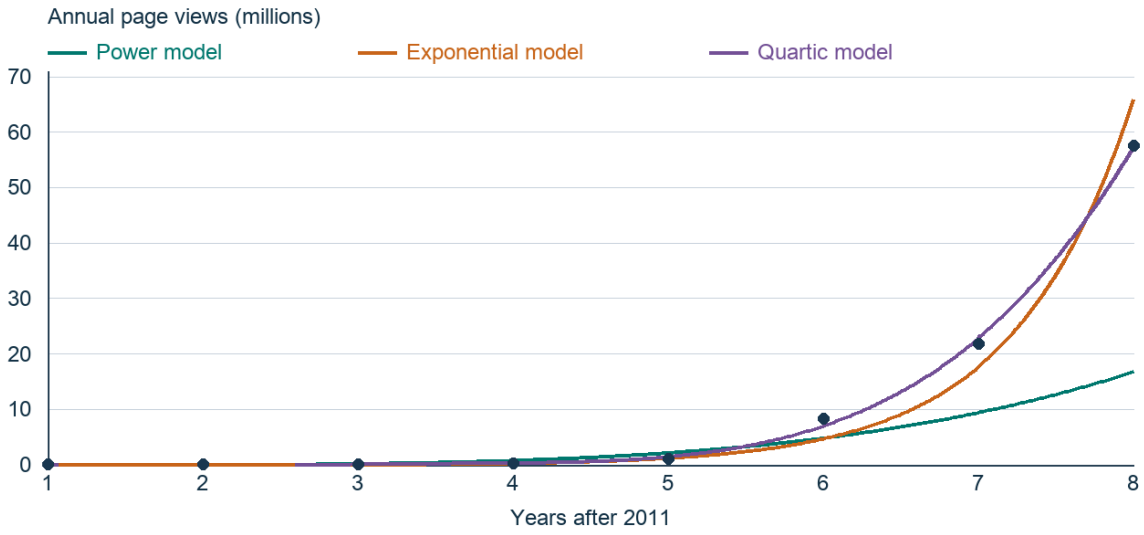
?

A website has had the following annual numbers of page views each year:

Year	2012	2013	2014	2015	2016	2017	2018	2019
Views **V**	10115	26790	60306	180386	1119801	8.3 m	21.9 m	57.5 m

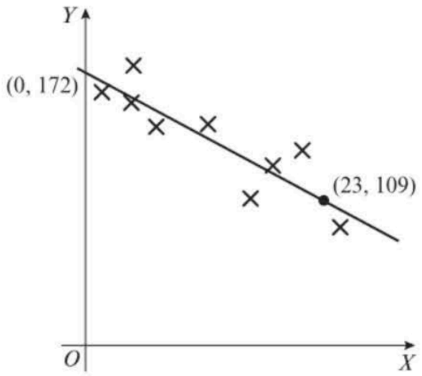
Determine the appropriate constants if we assume a power model $V = at^b$, where t is the number of years after 2011.

The power model underestimates the later years.
The exponential model follows the later growth more closely.



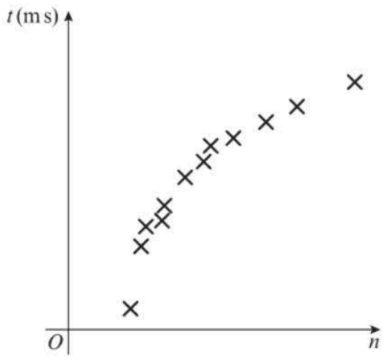
(P) **3** The scatter diagram shows the relationship between two sets of coded data, X and Y, where $X = \log x$ and $Y = \log y$. The regression line of Y on X is shown, and passes through the points (0, 172) and (23, 109).

The relationship between the original data sets is modelled by an equation of the form $y = ax^n$.
Find, correct to 3 decimal places, the values of a and n.



?

5 The time, t m s, needed for a computer algorithm to determine whether a number, n, is prime is recorded for different values of n. A scatter graph of t against n is drawn.



a Explain why a model of the form $t = a + bn$ is unlikely to fit these data.

The data are coded using the changes of variable $y = \log t$ and $x = \log n$. The regression line of y on x is found to be $y = -0.301 + 0.6x$.

b Find an equation for t in terms of n, giving your answer in the form $t = an^k$, where a and k are constants to be found.

a?

b?

6 Data are collected on the number of units (c) of a catalyst added to a chemical process, and the rate of reaction (r).

The data are coded using $x = \log c$ and $y = \log r$. It is found that a linear relationship exists between x and y and that the equation of the regression line of y on x is $y = 1.31x - 0.41$.

Use this equation to determine an expression for r in terms of c.

?

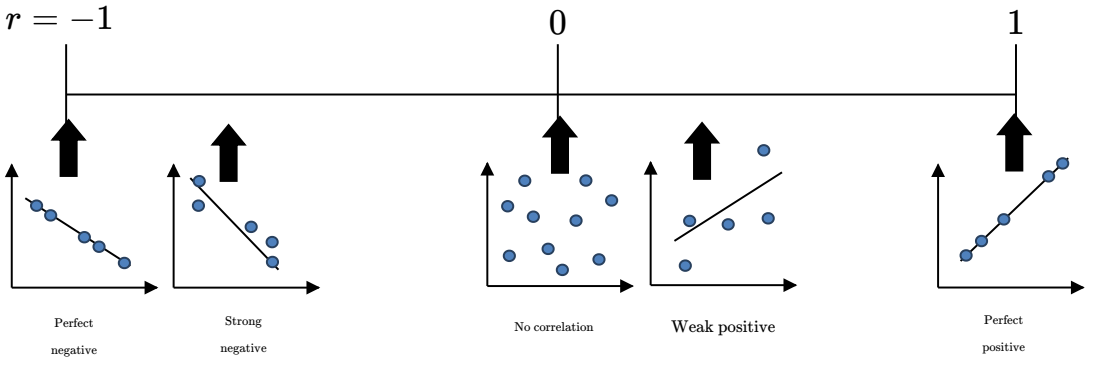
7 The heights, h cm, and masses, m kg, of a sample of Galapagos penguins are recorded. The data are coded using $y = \log m$ and $x = \log h$ and it is found that a linear relationship exists between x and y. The equation of the regression line of y on x is $y = 0.0023 + 1.8x$.

Find an equation to describe the relationship between m and h, giving your answer in the form $m = ah^n$, where a and n are constants to be found.

?

You're used to use qualitative terms such as "positive correlation" and "negative correlation" and "no correlation" to describe the **type** of correlation, and terms such as "perfect", "strong" and "weak" to describe the **strength**.
The **Product Moment Correlation Coefficient** is one way to quantify this:

The product moment correlation coefficient (PMCC), denoted by r, describes the linear correlation between two variables. It can take values between -1 and 1.



Rule of thumb: $r < -0.7$ or $r > 0.7$ is considered to be 'strong' correlation.

Note that PMCC is only applicable for a linear correlation, i.e. closeness of fit to a linear regression line (i.e. a straight 'line of best fit'). It may be the data exhibits strong correlation with respect to a different model (e.g. exponential) even when the PMCC is low.

Calculating r in the calculator

You must have a calculator that is capable of calculating r directly: in the A Level 2017+ syllabus you are no longer required to use formulae to calculate r.

x	y
1	3
2	6
3	5
4	8



$y = a + bx$

Data Entry

PMCC

The following instructions are for the Casio CG50.
Press MODE then select 'Statistics'.

We want to measure **linear** correlation, so select $y = a + bx$

Enter each of the x values in the table on the left.

to obtain r (i.e. the coefficients of your line of best fit and the PMCC), Press:
Graph (F1)
GRAPH1 (F1)
CALC (F1)
X (F2)
a+bx (F2)

You should obtain $\mathbf{r = 0.868}$

Example

[Textbook] From the large data set, the daily mean windspeed, w knots, and the daily maximum gust, g knots, were recorded for the first 10 days in September in Hurn in 1987.

Day of month	1	2	3	4	5	6	7	8	9	10
w	4	4	8	7	12	12	3	4	7	10
g	13	12	19	23	33	37	10	n/a	n/a	23

a. State the meaning of n/a in the table above.
b. Calculate the product moment correlation coefficient for the remaining 8 days.
c. With reference to your answer to part b, comment on the suitability of a linear regression model for these data.

a ?

b ?

c ?

This is a common exam question. The important bit is **evaluating the suitability of the chosen model** (in this case a linear regression model, i.e. line of best fit). The closer r is to 1 or to -1, the more suitable this linear regression model.

4 The number of atoms of a radioactive substance, n, is measured at various times, t minutes after the start of an experiment. The table below shows the data.

Time, t	1	2	4	5	7
Atoms, n	231	41	17	7	2
log n					

The data is coded using $x = t$ and $y = \log n$.

a Copy and complete the table showing the values of $\log n$. **(2 marks)**

b Calculate the product moment correlation coefficient for the coded data. **(1 mark)**

c With reference to your answer to **b**, state whether an exponential model is a good fit for these data. **(2 marks)**

The equation of the regression line of y on x is found to be $y = 2.487 - 0.320x$.

d Find an expression for n in terms of t, giving your answer in the form $n = ab^t$, where a and b are constants to be found. **(3 marks)**

a?

b?

c?

d?

5 The width, w cm, and the mass, m grams, of snowballs are measured. The table below shows the data.

Width, w	3	4	6	8	11
Mass, m	23	40	80	147	265
log w					
log m					

The data are coded using $x = \log w$ and $y = \log m$.

a Copy and complete the table showing the values of $\log w$ and $\log m$. **(3 marks)**

b Calculate the product moment correlation coefficient for the coded data. **(1 mark)**

c With reference to your answer to **b**, state whether a model in the form $y = kx^n$ where k and n are constants is a good fit for these data. **(2 marks)**

d Determine the values of k and n. **(3 marks)**

a?

b?

c?

d?

6 From the large data set, the daily mean air temperature, t °C, and the rainfall, f mm, were recorded for Perth on seven consecutive days in August 2015.

Temp, t	18.0	16.4	15.3	15.0	13.7	10.2	12.0
Rainfall, f	3.0	13.0	4.6	32.0	28.0	63.0	22.0

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a Calculate the product moment correlation coefficient for these data. **(1 mark)**

b With reference to your answer to part **a**, comment on the suitability of a linear regression model for these data. **(2 marks)**

a?

b?

7 From the large data set, the daily total rainfall, x mm, and the daily total sunshine, y hours, were recorded for Camborne on seven consecutive days in May 2015.

Rainfall, x	2.2	tr	1.4	4.4	tr	0.2	0.6
Sunshine, y	5.2	7.7	5.6	0.3	5.1	0.1	8.9

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a State the meaning of 'tr' in the table above. **(1 mark)**

b Calculate the product moment correlation coefficient for these 7 days, stating clearly how you deal with the entries marked 'tr'. **(2 marks)**

c With reference to your answer to part **b**, comment on the suitability of a linear regression model for these data. **(2 marks)**

a?

b?

c?

Data Recorded for two varilables x and y

x	3.1	5.6	7.1	8.6	9.4	10.7
y	3.2	4.8	5.7	6.5	6.9	7.6

By calculating the product moment correlation coefficients for suitability coded values of x and y state, with reasons, whether these data are more closely modelled by a relationship of the form $y = ab^x$ or a relationship of the form $y = kx^n$, where a, b, k and n are constants.

?

	B	C	D	E	G	H
1		English Exam Mark			Maths Exam Mark	
2		Mean	60		Mean	70
3	Student	S.D.	5		S.D.	10
4	1		63.90			70.13
5	2		55.24			65.99
6	3		58.80			80.18
7	4		59.65			57.16
8	5		66.44			72.76
9	6		59.53			79.82
10	7		57.43			71.48
11	8		58.33			60.56
12	9		67.43			69.56
13	10		63.11			87.13
16		r=	0.219			

	B	C	D	E	G	H
1		English Exam Mark			Maths Exam Mark	
2		Mean	60		Mean	70
3	Student	S.D.	5		S.D.	10
4	1		60.22			74.64
5	2		62.25			79.15
6	3		61.30			75.29
7	4		60.61			71.35
8	5		55.31			74.05
9	6		57.13			89.73
10	7		57.16			70.41
11	8		58.96			60.31
12	9		56.30			71.95
13	10		63.23			69.95
16		r=	-0.094			

Suppose we use a spreadsheet to randomly generate maths marks for students, and separately generate random English marks.

r denotes the PMCC of a sample.

What is the **observed** PMCC between Maths and English marks in this first set of data?

?

ρ (Greek letter *rho*) is the PMCC for the **whole population**.

But what is the true underlying PMCC between Maths and English?

?

$\therefore$ r is the test statistic, ρ is the population parameter.

	B	C	D	E	G	H
1		English Exam Mark			Maths Exam Mark	
2		Mean	60		Mean	70
3	Student	S.D.	5		S.D.	10
4	1		63.90			70.13
5	2		55.24			65.99
6	3		58.80			80.18
7	4		59.65			57.16
8	5		66.44			72.76
9	6		59.53			79.82
10	7		57.43			71.48
11	8		58.33			60.56
12	9		67.43			69.56
13	10		63.11			87.13
16		r=	0.219			

	B	C	D	E	G	H
1		English Exam Mark			Maths Exam Mark	
2		Mean	60		Mean	70
3	Student	S.D.	5		S.D.	10
4	1		60.22			74.64
5	2		62.25			79.15
6	3		61.30			75.29
7	4		60.61			71.35
8	5		55.31			74.05
9	6		57.13			89.73
10	7		57.16			70.41
11	8		58.96			60.31
12	9		56.30			71.95
13	10		63.23			69.95
16		r=	-0.094			

So despite that $\rho = 0$ (i.e. English/Maths marks have no underlying correlation), the observed PMCC for the sample, r, could vary, and might, purely by chance, suggest some correlation. In the first sample, this happened to be positive, in the latter, negative.

If we increased the sample size (in this case 10), we'd expect the observed PMCC to more closely match the true PMCC (0).

A natural question we might therefore ask is, if we had no knowledge about the underlying population: "**is the PMCC of $r = 0.219$ statistically significant, or are Maths/English marks not correlated, and this observed correlation emerged just by chance**?"

This sounds like a Hypothesis Test!

	B	C	D	E	G	H
1		English Exam Mark			Maths Exam Mark	
2		Mean	60		Mean	70
3	Student	S.D.	5		S.D.	10
4	1		63.90			70.13
5	2		55.24			65.99
6	3		58.80			80.18
7	4		59.65			57.16
8	5		66.44			72.76
9	6		59.53			79.82
10	7		57.43			71.48
11	8		58.33			60.56
12	9		67.43			69.56
13	10		63.11			87.13
14						
15						
16		r=	0.219			

CRITICAL VALUES FOR CORRELATION COEFFICIENTS

These tables concern tests of the hypothesis that a population correlation coefficient ρ is 0. The values in the tables are the minimum values which need to be reached by a sample correlation coefficient in order to be significant at the level shown, on a one-tailed test.

Product Moment Coefficient						Spearman's Coefficient		
		Level			Sample		Level	
0.10	0.05	0.025	0.01	0.005	Level	0.05	0.025	0.01
0.8000	0.9000	0.9500	0.9800	0.9900	4	1.0000	-	-
0.6870	0.8054	0.8783	0.9343	0.9587	5	0.9000	1.0000	1.0000
0.6084	0.7293	0.8114	0.8822	0.9172	6	0.8286	0.8857	0.9429
0.5509	0.6694	0.7545	0.8329	0.8745	7	0.7143	0.7857	0.8929
0.5067	0.6215	0.7067	0.7887	0.8343	8	0.6429	0.7381	0.8333
0.4716	0.5822	0.6664	0.7498	0.7977	9	0.6000	0.7000	0.7833
0.4428	0.5494	0.6319	0.7155	0.7646	10	0.5636	0.6485	0.7455
0.4187	0.5214	0.6021	0.6851	0.7348	11	0.5364	0.6182	0.7091
0.3981	0.4973	0.5760	0.6581	0.7079	12	0.5035	0.5874	0.6783
0.3802	0.4762	0.5529	0.6339	0.6835	13	0.4835	0.5604	0.6484
0.3646	0.4575	0.5324	0.6120	0.6614	14	0.4637	0.5385	0.6264

These values give the minimum value of r required to reject the null hypothesis, i.e. the amount of correlation that would be considered significant.

Let's carry out a hypothesis test on whether there is **positive** correlation between English and Maths marks, at 10% significance level:

H_0 : ?

There is no underlying correlation between English and maths marks.

H_1 : ?

There is an underlying positive correlation between English and maths marks.

Sample size

? **0**

Critical value for 10% significance level:

?

Look up value in table.

?

As with Year 1, a 2 mark conclusion: (a) Compare values; do we reject H_0? (b) put in context of original problem.

In the previous example we hypothesised that English/Maths marks were positively correlated. But we could also test whether there was **any** correlation, i.e. positive **or** negative.

[Textbook] A scientist takes 30 observations of the masses of two reactants in an experiment. She calculates a product moment correlation coefficient of $r = -0.45$.

The scientist believes there is no correlation between the masses of the two reactants. Test at the 10% level of significance, the scientist's claim, stating your hypotheses clearly.

Product Moment Coefficient					
		Level			Sample
0.10	0.05	0.025	0.01	0.005	size, n
0.8000	0.9000	0.9500	0.9800	0.9900	4
0.6870	0.8054	0.8783	0.9343	0.9587	5
0.6084	0.7293	0.8114	0.8822	0.9172	6
0.2992	0.3783	0.4438	0.5155	0.5614	20
0.2914	0.3687	0.4329	0.5034	0.5487	21
0.2841	0.3598	0.4227	0.4921	0.5368	22
0.2774	0.3515	0.4133	0.4815	0.5256	23
0.2711	0.3438	0.4044	0.4716	0.5151	24
0.2653	0.3365	0.3961	0.4622	0.5052	25
0.2598	0.3297	0.3882	0.4534	0.4958	26
0.2546	0.3233	0.3809	0.4451	0.4869	27
0.2497	0.3172	0.3739	0.4372	0.4785	28
0.2451	0.3115	0.3673	0.4297	0.4705	29
0.2407	0.3061	0.3610	0.4226	0.4629	30
0.2070	0.2638	0.3120	0.3665	0.4026	40
0.1843	0.2353	0.2787	0.3281	0.3610	50
0.1678	0.2144	0.2542	0.2997	0.3301	60

H_0 ?

H_1 ?

Sample size $= 3$?

Two-tailed, so 5% at each tail.

Critical value at ? significance:

?

?

Test your understanding 25

[Textbook] The table from the large data set shows the daily maximum gust, x kn, and the daily maximum relative humidity, y%, in Leeming for a sample of eight days in May 2015.

x	31	28	38	37	18	17	21	29
y	99	94	87	80	80	89	84	86

a. Find the product moment correlation coefficient for this data.

b. Test, at the 10% level of significance, whether there is evidence of a positive correlation between daily maximum gust and daily maximum relative humidity. State your hypotheses clearly.

a ?

b ?

Exercise 1C – Statistics Year 2 Pages 10-12 26

3 a State what is measured by the product moment correlation coefficient. **(1 mark)**

Twelve students sat two Biology tests, one theoretical the other practical. Their marks are shown below.

Marks in theoretical test, *t*	5	9	7	11	20	4	6	17	12	10	15	16
Marks in practical test, *p*	6	8	9	13	20	9	8	17	14	8	17	18

b Find the product moment correlation coefficient for these data, correct to 3 significant figures. **(2 marks)**

A teacher claims that students who do well in their theoretical test also tend to do well in their practical test.

c Test this claim at a 0.05 significance level, stating your hypotheses clearly. **(3 marks)**

d Give an interpretation of the value 0.05 in your hypothesis test. **(1 mark)**

a?

b?

c?

d?

4 The following table shows the marks attained by 8 students in English and Mathematics tests.

Student	A	B	C	D	E	F	G	H
English	25	18	32	27	21	35	28	30
Mathematics	16	11	20	17	15	26	32	20

a Calculate the product moment correlation coefficient. **(1 mark)**

b Test, at the 5% significance level, whether these results show evidence of a linear relationship between English and Mathematics marks. State your hypotheses clearly. **(3 marks)**

a?

b?

6 A scientific researcher collects data on the amount of solvent in a solution and the rate of reaction. She calculates the product moment correlation coefficient between the two sets of data and finds it to be –0.43. Given that she collected data from 15 samples, test, at the 5% level of significance, the claim that there is a negative correlation between the amount of solvent and the rate of reaction. State your hypotheses clearly. **(3 marks)**

?

9 An employee at a weather centre believes that there is a negative correlation between humidity and visibility. She takes a sample of data from Heathrow in August 1987.

Humidity (%)	92	93	91	82	91	100
Visibility (m)	2500	1500	2700	2900	2200	1000

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a Calculate the product moment correlation coefficient for these data. **(1 mark)**

b Test, at the 1% level of significance, the employee's claim. State your hypotheses clearly. **(3 marks)**

a?

b?